O Adapt Business and Technical Tradeoff Analysis

DING SHIYIN_ G2604052H

Problem and intended value

I built O-Adapt for recreational orienteers who want varied individual practice. I estimated that traditional preparation takes about three hours for course design and equipment placement but have not validated this through user research. A user requests distance, difficulty and training focus; the system checks spacing, connectivity and recorded evidence.

Orienteer is the closest existing tool I identified. It supports planning, tracking, and sharing courses, but its documented workflow still requires manual checkpoint placement. O-Adapt automatically produces a constraint-checked route from the request, RAG evidence, and a verified graph for one fixed park.

Limiting the MVP to Jurong Central Park reduces geographic coverage but makes field checking and reproducible evaluation feasible within the project period.

Technology choice and baseline

O-Adapt is a hybrid system. A deterministic graph engine enumerates up to twelve candidates that pass connectivity, distance, and spacing gates. Explicit retrieval selects relevant entries from eight curated rules. A rented foundation model uses the request and those rules to choose the final candidate. Server code rejects unknown model IDs and derives geometry and verification status from held data.

I did not use an agent because bounded selection needs no tool loop, and an agent would add unmeasured cost and failure modes.

Existing control mode uses identifiable features from the verified set. Exploration mode creates virtual GPxx GPS controls on path-network nodes that I walked and verified; it checks distance, spacing and recorded hazards without requiring physical flags or distinctive features.

The non-AI baseline is the engine's highest-ranked valid candidate. Giving both methods the same candidates isolate the model's effect without letting it override validation. In the live run, 15/17 model choices differed from the baseline. Route summaries remain templated because five earlier cases scored the template 5.0/5 versus 4.2/5 for model text. I did not use supervised learning because I lack labels, or a vector database because eight knowledge entries are clearer with rule-and-tag retrieval.

Build versus buy

Layer	Decision and reason
Interface and serving	Own the HTML/SVG interface and Node server for inspectable behavior.
Orchestration	Own the one-request flow and validation logic.
Model	Rent openai/gpt-4o-mini through OpenRouter to avoid training and hosting costs.
Data and retrieval	Own the curated corpus and local graph processing; retain OSM attribution.
Evaluation and observability	Own fixtures, regression checks and JSON logs.

I chose code so browser and server share testable routing constraints; I did not evaluate a low-code product. Deployment requires Node 20+, the repository, and an API key. Model failure is explicit.

Cost and latency

The model path makes one capped call per course; the baseline makes none. The final live batch completed 17/17 selections using 37,750 tokens at a provider-reported cost of US$0.00602205 and 1.74 seconds average latency. A

30-second timeout causes failure. Evaluation stops after reported spending reaches $1 or becomes unknown; a restricted provider key is needed for a hard cap.

Data and measured implementation

The graph contains 288 OSM-derived segments: 238 marked field-verified and 50 excluded, plus fourteen eligible controls and eight knowledge entries. I walked and checked eligible segments on 23 September 2026. Fourteen photos I supplied document the controls and visible approaches. OSM attribution is recorded.

The offline suite contains twenty normal requests and ten edge cases. Seventeen requests produced verified courses and three abstained, giving an 85% yield and meeting the 16/20 target. All thirty behavioural contracts passed. Six normal exploration requests produced constraint-valid virtual-control routes on verified path nodes; four exploration edge cases abstained, so exploration achieved 10/10 contract compliance.

A published twenty-row binary checklist reports path, control, spacing, +/-10% distance and hazard results separately. It distinguishes a complete usable route from a correct safe abstention.

The suite covers exclusions, missing verification, hazards, missing policies, and invalid inputs. An independent haversine calculation checks all-pairs spacing but does not prove current physical conditions or recompute every graph of distance. A separate twenty-request set was generated by Gemini 2.5 Flash and frozen before evaluation; it has no human feasibility labels.

All 17 original live model-selected routes passed deterministic contracts. On the new set, 18/20 requests yielded routes, and all 18 passed recorded constraints. Gemini blindly preferred the model route in 3/10 differing pairs, the baseline in 4/10, and tied 3/10. Five human-reviewed routes averaged 4.60/5 request fit and 4.40/5 focus fit; N03 exposed weak advanced-navigation classification. On those five, Gemini's focus scores differed from human scores by 1.20 points on average. Model judging does not establish better route quality.

Outcome critique, tuning and future path

The model selects valid routes while deterministic code controls geometry, verification, and abstention. The 85% yield meets my target, but three abstentions show that a fixed control set cannot satisfy every request. Exploration adds virtual GPS controls to the verified network without flags or distinctive features. The model was inexpensive and reliable, but its training-quality benefit remains unproven.

The hardest engineering issue was giving the model influence without authority over verification of claims. I moved enumeration into one shared engine, sent only validated candidates, required an exact candidate ID, and derived metrics from server-held data. I also removed an ambiguous control and separated existing physical-feature controls from generated virtual GPS controls, preventing invented controls or altered metrics from appearing approved.

Nearest-node connectors still approximate access; feature labels are weak proxies for training quality, and field records can become stale. Future work should measure route diversity, attach dated evidence to individual paths, obtain human-labelled holdout cases, time manual versus assisted preparation, and compare selections with the baseline through blinded human reviews.

Optional browser tracking records ordered arrivals locally.

Risks, mitigations and limits

A silent failure is a numerically valid route that poorly serves the requested training focus. The model cannot alter candidates or verification, and the interface displays engine-controlled checks, but human review is still needed for training quality. Unverified edges and blocked controls are excluded. GPxx points are virtual GPS waypoints on verified path-network nodes, so no physical control placement is required; GPS drift and temporary site changes remain relevant. Outages are visibly labeled.

Residual risks include stale records, inaccurate coordinates, approximate connectors and difficulty heuristics that do not prove genuine navigation challenges. Existing-control results apply only to the recorded dataset, so users must check current conditions. I claim neither certified mapping nor safety, and the local server is not a production multi-user deployment.

Location data stays in browser storage and can be exported or deleted; any shared service would need consent and access controls.